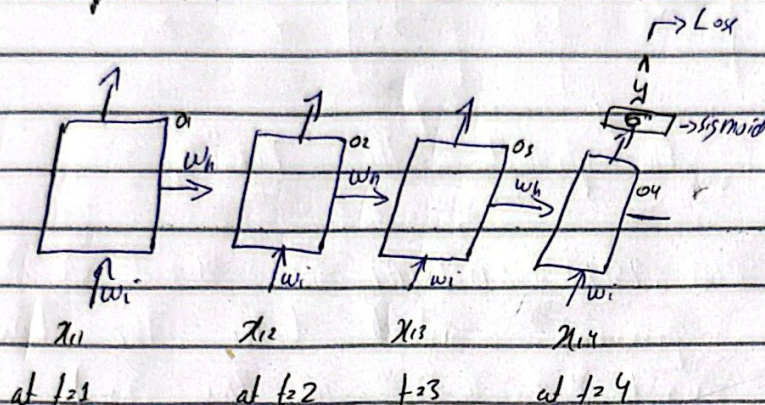


⇒ Bidirectional RNN :

⇒ in Single RNN or LSTM :

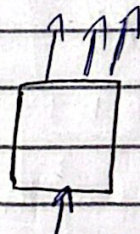


⇒ Types of RNN:

- One to Many RNN { 1 input, Many outputs }
- Many to One RNN
- Many to Many RNN
- One to One RNN { 1 input, 1 output }

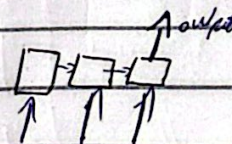


• one to one



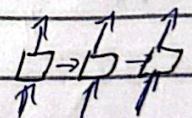
• one to Many

Image Captioning.
give 1 img to Model & it give text



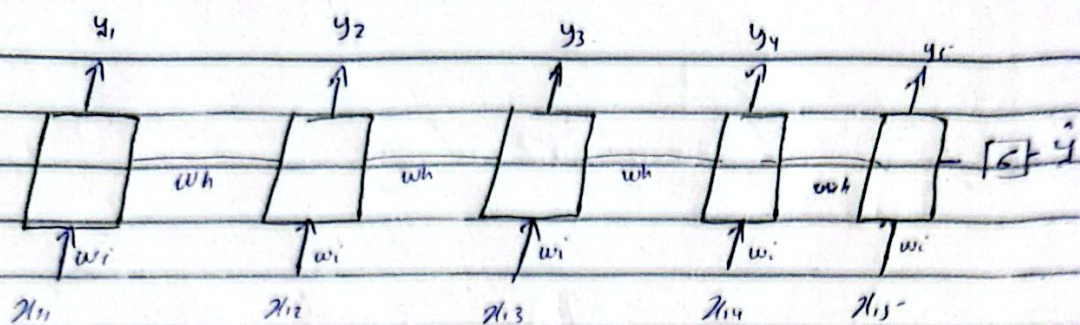
• Many to One

I Love this Movie
↓
RNN
↓
Positive



• Many to Many

Language Translation
French → Le Pizzeria
↓
Eng → Love Pizzeria



Text Example: Alyan eats — in Hotel.
 x_{11} x_{12} x_{13} x_{14} x_{15}

* I need to predict the " x_{13} ".

↳ in simple RNN, LSTM, GRU, LSTM $\Rightarrow x_{13}$ is dependent on its previous content (The the two previous word " x_{11} & x_{12} " in This case).

* But here " x_{13} " is also dependent of on upcoming context (x_{14} , x_{15}).

* I need to predict " y_3 " for that I need previous content & forward context too.

* This is Achieved Through Bidirectional RNN.

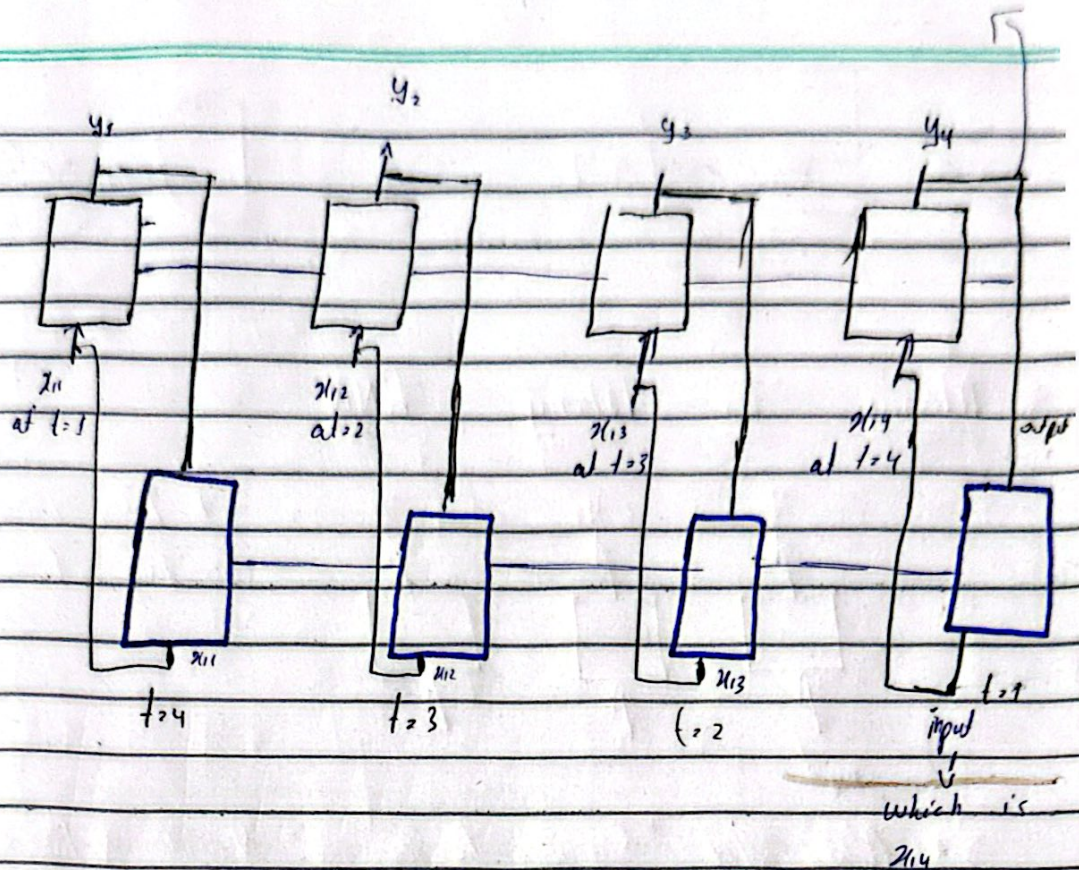
*** We Make another RNN & Process the sentence from Reverse direction.

→ Connect this RNN with Normal RNN.

→ Now it has context of forward words too.

I need to predict y_3

Combine output of both



Allym eats — in H.O./ed.
 x_{11} x_{12} x_{13} x_{14} x_{15}

Now to predict x_{13} I will be having the forward context too.

We will have context of both previous & upcoming words.